

Congruence of Linear Symplectic Forms by the Symplectic Group

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Abstract

We study the congruence action of the real symplectic group on nondegenerate skew-symmetric bilinear forms. Fix the standard symplectic matrix J_n on $\mathbb{R}^{2n}$ and let

$$\mathcal{S}_{2n} = \{A \in M_{2n}(\mathbb{R}) : A^T = -A, \det A \neq 0\}.$$

The group $\mathrm{Sp}(2n, \mathbb{R}) = \{P : P^T J_n P = J_n\}$ acts on $\mathcal{S}_{2n}$ by $A \mapsto P^T A P$. We first prove that the Pfaffian and the linear invariant

$$s(A) = \sum_{i=1}^n A_{2i-1, 2i} = -\frac{1}{2} \mathrm{tr}(J_n A)$$

are invariant under this action. In dimension four, these invariants give a complete classification away from the fixed scalar forms cJ_2 : two non-scalar forms lie in the same orbit if and only if they have the same Pfaffian and the same value of s . We also describe the topology of the regular orbit level sets in dimension four. In higher dimensions, we construct n invariant polynomial functions: the coefficients of $\mathrm{Pf}(tJ_n + A)$. Finally, we explain how these linear invariants give pointwise obstructions to equivalence of nonlinear symplectic forms under symplectomorphisms of $(\mathbb{R}^{2n}, \omega_{\mathrm{std}})$, and derive global invariants for split diagonal families.

Keywords: Symplectic Forms, Symplectomorphisms, Invariants, Orbit Space, Symplectic Geometry

1 Introduction

1.1 Motivation

Symplectic structures on $\mathbb{R}^{2n}$ have been studied extensively under equivalence by diffeomorphisms. In this paper, we restrict the equivalence relation: we fix the standard symplectic form ω_{std} and ask how its symplectomorphism group acts on other symplectic forms. The linear version of this question is the congruence action

$$A \longmapsto P^T A P, \quad P \in \mathrm{Sp}(2n, \mathbb{R}),$$

on nondegenerate skew-symmetric matrices. The invariants obtained from this linear action give pointwise obstructions to equivalence of nonlinear symplectic forms by ω_{std} -symplectomorphisms.

The main correction made in this version is to state the invariants and their normalizations consistently, and to separate the dimension-four classification from the higher-dimensional coefficient invariants. In dimension four, the Pfaffian and the sum function classify all non-scalar orbits. In higher dimensions, the coefficients of $\mathrm{Pf}(tJ_n + A)$ remain invariants, but this cleaned version treats their role as invariants rather than as a complete classification theorem.

1.2 Goals and notation

Throughout, all vectors and matrices are written with respect to the standard basis. Let

$$J_0 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad J_n = \text{diag}(J_0, \dots, J_0).$$

Set

$$\mathcal{S}_{2n} = \{A \in M_{2n}(\mathbb{R}) : A^T = -A, \det A \neq 0\}, \quad \text{Sp}(2n, \mathbb{R}) = \{P \in M_{2n}(\mathbb{R}) : P^T J_n P = J_n\}.$$

We study the action

$$\rho : \text{Sp}(2n, \mathbb{R}) \times \mathcal{S}_{2n} \rightarrow \mathcal{S}_{2n}, \quad \rho(P, A) = P^T A P.$$

When no confusion is possible, we write $\text{Sp}(2n)$ for $\text{Sp}(2n, \mathbb{R})$.

The paper addresses the following questions. First, which scalar functions of A are invariant under this action? Second, in dimension four, do the Pfaffian and sum function classify the orbit space? Third, what do the corresponding orbit level sets look like topologically? Finally, how do the linear invariants produce pointwise and global obstructions in the nonlinear setting?

2 Main results

2.1 The four-dimensional classification

For

$$A = \begin{pmatrix} 0 & a & b & c \\ -a & 0 & d & e \\ -b & -d & 0 & f \\ -c & -e & -f & 0 \end{pmatrix} \in \mathcal{S}_4,$$

define

$$\text{Pf}(A) = af - be + cd, \quad s(A) = a + f.$$

Equivalently, $s(A) = -\frac{1}{2} \text{tr}(J_2 A)$.

Theorem 2.1 (Four-dimensional invariants). *For every $P \in \text{Sp}(4)$ and $A \in \mathcal{S}_4$,*

$$\text{Pf}(P^T A P) = \text{Pf}(A), \quad s(P^T A P) = s(A).$$

Scalar multiples of J_2 are fixed by the action: $P^T(cJ_2)P = cJ_2$ for all $P \in \text{Sp}(4)$ and $c \in \mathbb{R}^*$. The remaining orbits are classified by the pair (Pf, s) .

Theorem 2.2 (Orbit classification in dimension four). *The orbits of $\mathcal{S}_4$ under $\text{Sp}(4)$ are as follows.*

1. *For each $c \in \mathbb{R}^*$, the singleton $\{cJ_2\}$ is an orbit.*
2. *For each $(p, q) \in \mathbb{R}^* \times \mathbb{R}$, the set*

$$\mathcal{O}_{p,q} = \{A \in \mathcal{S}_4 : \text{Pf}(A) = p, s(A) = q, A \notin \mathbb{R}J_2\}$$

is either empty or a single orbit.

In particular, if $A, B \in \mathcal{S}_4 \setminus \mathbb{R}J_2$, then A and B lie in the same $\text{Sp}(4)$ -orbit if and only if

$$\text{Pf}(A) = \text{Pf}(B), \quad s(A) = s(B).$$

2.2 Topology of the four-dimensional level sets

For $p \in \mathbb{R}^*$ and $q \in \mathbb{R}$, define the level set

$$L_{p,q} = \{(a, b, c, d, e, f) \in \mathbb{R}^6 : af - be + cd = p, a + f = q\}$$

and set

$$\Delta = \frac{q^2}{4} - p.$$

Theorem 2.3 (Regular orbit-level topology). *If $\Delta > 0$, then*

$$L_{p,q} \cong S^2 \times \mathbb{R}^2.$$

If $\Delta < 0$, then

$$L_{p,q} \cong (\mathbb{R}^2 \setminus \{0\}) \times \mathbb{R}^2.$$

Remark 2.4. The case $\Delta = 0$ should be stated and proved separately. This is also the case in which the scalar fixed points $\pm\sqrt{p}J_2$ can occur when $p > 0$.

2.3 Higher-dimensional invariants

We use the standard normalization

$$\text{Pf}(A) = \frac{1}{2^n n!} \sum_{\sigma \in S_{2n}} \text{sgn}(\sigma) \prod_{i=1}^n A_{\sigma(2i-1), \sigma(2i)}.$$

Equivalently, one may sum over unordered pairings with the usual ordering convention.

For $A \in \mathcal{S}_{2n}$, define $s_k(A)$ by the polynomial identity

$$\text{Pf}(tJ_n + A) = \sum_{k=0}^n s_k(A) t^k,$$

so that $s_n(A) = 1$ and $s_0(A) = \text{Pf}(A)$. Equivalently, for $0 \leq k \leq n-1$,

$$s_k(A) = \sum_{\substack{S \subset \{1, \dots, n\} \\ |S|=k}} \text{Pf}(A_S),$$

where A_S is obtained from A by deleting rows and columns $2i-1, 2i$ for every $i \in S$.

Theorem 2.5 (Higher-dimensional linear invariants). *For every $A \in \mathcal{S}_{2n}$, $P \in \text{Sp}(2n)$, and $0 \leq k \leq n$,*

$$s_k(P^T A P) = s_k(A).$$

2.4 Nonlinear consequences

Let ω be a symplectic form on $\mathbb{R}^{2n}$ and let $\Omega(x)$ denote its matrix in the standard coordinates. If $\phi \in \text{Symp}(\mathbb{R}^{2n}, \omega_{\text{std}})$ and $\phi^* \omega_1 = \omega_2$, then

$$(d\phi_x)^T \Omega_1(\phi(x)) d\phi_x = \Omega_2(x),$$

with $d\phi_x \in \text{Sp}(2n)$.

Theorem 2.6 (Pointwise obstruction). *If $\phi^*\omega_1 = \omega_2$ for some $\phi \in \text{Symp}(\mathbb{R}^{2n}, \omega_{\text{std}})$, then for every $x \in \mathbb{R}^{2n}$ and every $0 \leq k \leq n$,*

$$s_k(\Omega_1(\phi(x))) = s_k(\Omega_2(x)).$$

For split diagonal symplectic forms

$$\omega_f = \sum_{i=1}^n f_i(x_i, y_i) dx_i \wedge dy_i, \quad \omega_g = \sum_{i=1}^n g_i(x_i, y_i) dx_i \wedge dy_i,$$

where all f_i, g_i are nowhere vanishing, Theorem 2.6 implies a pointwise multiset equality.

Corollary 2.7 (Split diagonal forms). *Suppose $\phi^*\omega_f = \omega_g$ for some $\phi \in \text{Symp}(\mathbb{R}^{2n}, \omega_{\text{std}})$, where $\phi = (\phi_1, \dots, \phi_{2n})$. Then, for every $x \in \mathbb{R}^{2n}$,*

$$\{f_i(\phi_{2i-1}(x), \phi_{2i}(x)) : 1 \leq i \leq n\} = \{g_i(x_i, y_i) : 1 \leq i \leq n\}$$

as multisets.

Consequently, if

$$m_f(x) = \min_{1 \leq i \leq n} f_i(x_i, y_i), \quad M_f(x) = \max_{1 \leq i \leq n} f_i(x_i, y_i),$$

then the quantities

$$\inf_x m_f(x), \quad \sup_x m_f(x), \quad \inf_x M_f(x), \quad \sup_x M_f(x)$$

are invariants of the split diagonal form under ω_{std} -symplectomorphism.

3 Linear invariants

We prove the invariance statements in a form valid in every even dimension. Recall that

$$\mathcal{S}_{2n} = \{A \in M_{2n}(\mathbb{R}) : A^T = -A, \det A \neq 0\}, \quad \text{Sp}(2n) = \{P : P^T J_n P = J_n\}.$$

3.1 The Pfaffian and the trace invariant

We use two standard facts. First, for every skew-symmetric A and every invertible P ,

$$\text{Pf}(P^T A P) = \det(P) \text{Pf}(A). \tag{1}$$

Second, every real symplectic matrix has determinant one. Hence, if $P \in \text{Sp}(2n)$, then

$$\text{Pf}(P^T A P) = \text{Pf}(A).$$

The second invariant is even simpler. Define

$$s(A) = \sum_{i=1}^n A_{2i-1, 2i} = -\frac{1}{2} \text{tr}(J_n A).$$

If $P^T J_n P = J_n$, then also $P J_n P^T = J_n$. Therefore, using cyclicity of trace,

$$\begin{aligned} s(P^T A P) &= -\frac{1}{2} \text{tr}(J_n P^T A P) \\ &= -\frac{1}{2} \text{tr}(P J_n P^T A) \\ &= -\frac{1}{2} \text{tr}(J_n A) \\ &= s(A). \end{aligned}$$

This proves the invariance of both Pf and s .

Remark 3.1. This trace proof replaces the longer component-wise basis proof. It also makes clear why the invariant called the “sum function” is basis-dependent but natural relative to the fixed standard symplectic form J_n .

3.2 The coefficient invariants

Define $s_0, \dots, s_n$ by

$$\text{Pf}(tJ_n + A) = \sum_{k=0}^n s_k(A)t^k. \quad (2)$$

Then $s_n(A) = \text{Pf}(J_n) = 1$ and $s_0(A) = \text{Pf}(A)$. Moreover, $s_{n-1}(A) = s(A)$.

Proposition 3.2. *For every $P \in \text{Sp}(2n)$ and every $A \in \mathcal{S}_{2n}$,*

$$s_k(P^T A P) = s_k(A), \quad 0 \leq k \leq n.$$

Proof. Since $P^T J_n P = J_n$ and $\det(P) = 1$, the congruence identity for the Pfaffian gives

$$\begin{aligned} \text{Pf}(tJ_n + P^T A P) &= \text{Pf}(P^T (tJ_n + A) P) \\ &= \det(P) \text{Pf}(tJ_n + A) \\ &= \text{Pf}(tJ_n + A). \end{aligned}$$

The two sides are polynomials in t , so their coefficients are equal. □

3.3 Equivalent deletion formula

For $S \subset \{1, \dots, n\}$, let A_S denote the matrix obtained from A by deleting rows and columns $2i-1, 2i$ for every $i \in S$. Then, for $0 \leq k \leq n-1$,

$$s_k(A) = \sum_{\substack{S \subset \{1, \dots, n\} \\ |S|=k}} \text{Pf}(A_S). \quad (3)$$

Indeed, the coefficient of t^k in $\text{Pf}(tJ_n + A)$ is obtained by choosing the tJ_0 block in exactly k of the standard symplectic coordinate pairs and using A on the remaining coordinate pairs. This gives exactly the deletion formula above.

Remark 3.3. If the deletion formula is used as the definition of s_k , then Proposition 3.2 proves its invariance after identifying it with the coefficient formula (2). The coefficient definition is shorter and less error-prone.

4 Nonlinear consequences

Let ω be a symplectic form on $\mathbb{R}^{2n}$ and write $\Omega(x)$ for its matrix in the standard coordinates, so that

$$\omega(x)(v, w) = v^T \Omega(x) w.$$

Let ω_{std} denote the standard symplectic form with matrix J_n .

Proposition 4.1 (Pointwise linear obstruction). *Let ω_1 and ω_2 be symplectic forms on $\mathbb{R}^{2n}$. Suppose*

$$\phi \in \text{Symp}(\mathbb{R}^{2n}, \omega_{\text{std}}) \quad \text{and} \quad \phi^* \omega_1 = \omega_2.$$

Then, for every $x \in \mathbb{R}^{2n}$ and every $0 \leq k \leq n$,

$$s_k(\Omega_1(\phi(x))) = s_k(\Omega_2(x)).$$

Proof. The identity $\phi^* \omega_1 = \omega_2$ gives

$$(d\phi_x)^T \Omega_1(\phi(x)) d\phi_x = \Omega_2(x).$$

Since ϕ preserves ω_{std} , its derivative satisfies

$$(d\phi_x)^T J_n(d\phi_x) = J_n,$$

so $d\phi_x \in \text{Sp}(2n)$. The result follows from the linear invariance of the coefficients s_k . $\square$

4.1 Split diagonal forms

Now restrict to symplectic forms of the form

$$\omega_f = \sum_{i=1}^n f_i(x_i, y_i) dx_i \wedge dy_i, \quad \omega_g = \sum_{i=1}^n g_i(x_i, y_i) dx_i \wedge dy_i,$$

where the functions f_i, g_i are smooth and nowhere vanishing. Their matrix representations are block diagonal with i th block $f_i J_0$ or $g_i J_0$.

Proposition 4.2 (Pointwise multiset invariant). *Suppose $\phi^* \omega_f = \omega_g$ for some $\phi \in \text{Symp}(\mathbb{R}^{2n}, \omega_{\text{std}})$, where $\phi = (\phi_1, \dots, \phi_{2n})$. Then, for every $x = (x_1, y_1, \dots, x_n, y_n)$,*

$$\{f_i(\phi_{2i-1}(x), \phi_{2i}(x)) : 1 \leq i \leq n\} = \{g_i(x_i, y_i) : 1 \leq i \leq n\}$$

as multisets.

Proof. For a split diagonal form, the polynomial $\text{Pf}(tJ_n + \Omega(x))$ factors as

$$\prod_{i=1}^n (t + f_i(x_i, y_i)).$$

By Proposition 4.1, this polynomial is the same for $\Omega_f(\phi(x))$ and $\Omega_g(x)$ at every point x . Equality of monic polynomials implies equality of their roots, counted with multiplicity, which gives the asserted multiset equality. $\square$

Corollary 4.3 (Global invariants for split diagonal forms). *For $x = (x_1, y_1, \dots, x_n, y_n)$, define*

$$m_f(x) = \min_{1 \leq i \leq n} f_i(x_i, y_i), \quad M_f(x) = \max_{1 \leq i \leq n} f_i(x_i, y_i).$$

If ω_f and ω_g are ω_{std} -symplectomorphic, then

$$\begin{aligned} \inf_{x \in \mathbb{R}^{2n}} m_f(x) &= \inf_{x \in \mathbb{R}^{2n}} m_g(x), & \sup_{x \in \mathbb{R}^{2n}} m_f(x) &= \sup_{x \in \mathbb{R}^{2n}} m_g(x), \\ \inf_{x \in \mathbb{R}^{2n}} M_f(x) &= \inf_{x \in \mathbb{R}^{2n}} M_g(x), & \sup_{x \in \mathbb{R}^{2n}} M_f(x) &= \sup_{x \in \mathbb{R}^{2n}} M_g(x). \end{aligned}$$

Proof. By Proposition 4.2, for every x ,

$$m_f(\phi(x)) = m_g(x), \quad M_f(\phi(x)) = M_g(x).$$

Since ϕ is a bijection, the ranges of m_f and m_g agree, and the ranges of M_f and M_g agree. Taking infima and suprema gives the result. $\square$

4.2 A family that is diffeomorphic but not ω_{std} -symplectomorphic

Let $g : \mathbb{R}^2 \rightarrow \mathbb{R}$ be smooth with $\inf g = c > 0$, and define

$$\omega_t = (g(x_1, y_1) + t) dx_1 \wedge dy_1 + (g(x_2, y_2) + t) dx_2 \wedge dy_2, \quad t \in [0, 1].$$

Under the hypotheses of the Curry–Pelayo–Tang stability theorem, these forms are mutually diffeomorphic. However, if $t_1 > t_2$, then ω_{t_1} and ω_{t_2} are not ω_{std} -symplectomorphic.

Indeed, if $\phi^* \omega_{t_1} = \omega_{t_2}$, Proposition 4.2 would imply, for every x , that

$$g(\phi_1(x), \phi_2(x)) + t_1$$

equals either $g(x_1, y_1) + t_2$ or $g(x_2, y_2) + t_2$. Choose u, v with $g(u, v) < c + (t_1 - t_2)$ and set $x = (u, v, u, v)$. Then the right-hand side equals $g(u, v) + t_2$, while the left-hand side is at least $c + t_1$, a contradiction. Thus no such ϕ exists.

In particular, taking $g(x, y) = x^2 + y^2 + 1$ gives the explicit family

$$\omega_t = (x_1^2 + y_1^2 + t + 1) dx_1 \wedge dy_1 + (x_2^2 + y_2^2 + t + 1) dx_2 \wedge dy_2.$$

5 Further directions

We close with several questions suggested by the preceding results.

5.1 Higher-dimensional orbit classification

The dimension-four classification shows that, away from the scalar fixed points, the two invariants Pf and s classify $\text{Sp}(4)$ -orbits in $\mathcal{S}_4$. In higher dimensions, Theorem 2.5 gives n invariant coefficients $s_0, \dots, s_{n-1}$, where

$$\text{Pf}(tJ_n + A) = t^n + s_{n-1}(A)t^{n-1} + \dots + s_0(A).$$

Question 5.1. *Do the invariants $s_0, \dots, s_{n-1}$ classify the non-scalar $\text{Sp}(2n)$ -orbits in $\mathcal{S}_{2n}$? Equivalently, if $A, B \in \mathcal{S}_{2n} \setminus \mathbb{R}J_n$ and*

$$s_k(A) = s_k(B), \quad 0 \leq k \leq n-1,$$

must there exist $P \in \text{Sp}(2n)$ such that $B = P^T A P$?

This question is open in the present paper. The dimension-six case already requires care: scalar forms satisfy $\text{Pf}(cJ_3) = c^3$, so the scalar fixed-point family is parametrized by $c \in \mathbb{R}^*$, not by $\pm\sqrt{p}$ as in dimension four.

5.2 The singular orbit-geometry case

Theorem 2.3 treats the regular cases $\Delta \neq 0$ for the level sets

$$L_{p,q} = \{(a, b, c, d, e, f) : af - be + cd = p, a + f = q\}, \quad \Delta = \frac{q^2}{4} - p.$$

Question 5.2. *What is the topology of $L_{p,q}$ when $\Delta = 0$? How does this topology change after removing the scalar fixed point when $p > 0$ and $q = \pm 2\sqrt{p}$?*

5.3 Nonlinear symplectic forms

The pointwise invariants of Section 4 obstruct equivalence under ω_{std} -symplectomorphisms. They do not by themselves give sufficient conditions for equivalence.

Question 5.3. *For split diagonal families*

$$\omega_t = \sum_{i=1}^n f_i(t, x_i, y_i) dx_i \wedge dy_i, \quad t \in [0, 1],$$

what additional hypotheses ensure the existence of a smooth path

$$\phi_t \in \text{Symp}(\mathbb{R}^{2n}, \omega_{\text{std}}), \quad \phi_t^* \omega_t = \omega_0?$$

A natural first case is $n = 2$, where the pointwise invariants reduce to equality of the Pfaffian and trace invariant of the matrix representations. Understanding whether these pointwise constraints can be integrated to a global symplectomorphism remains an interesting problem.

References

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